

Homework 5

Due at 11:59 pm, May 25th

Please note that you must derive the answers to the first five questions manually, even though you are encouraged to use MPT3 to double-check your answers.

1 Set Operations (20 points)

Consider the following two sets in $\mathbb{R}^2$:

$$P = \{x \in \mathbb{R}^2 : -1 \leq x_1 \leq 2, -1 \leq x_2 \leq 1\},$$

and

$$Q = \{x \in \mathbb{R}^2 : |x_1| + |x_2| \leq 1\}.$$

Please answer the following questions:

- (a) Sketch P and Q .
- (b) Compute and sketch $P \cap Q$.
- (c) Describe geometrically the Minkowski sum

$$P \oplus Q = \{p + q : p \in P, q \in Q\}.$$

You do not need to give an irredundant half-space representation, but you should clearly indicate the resulting shape.

2 Image and Preimage of a Polyhedron Under an Affine Map (20 points)

Let

$$X = \{x \in \mathbb{R}^2 : -1 \leq x_1 \leq 1, -1 \leq x_2 \leq 1\},$$

and consider the affine map

$$y = Ax + b,$$

where

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Please answer the following questions:

- (a) Compute the image

$$Y = \{Ax + b : x \in X\}.$$

Sketch Y .

- (b) Let

$$S = \{y \in \mathbb{R}^2 : 0 \leq y_1 \leq 2, -1 \leq y_2 \leq 1\}.$$

Compute the preimage

$$\{x : Ax + b \in S\}.$$

Give the answer as linear inequalities in x_1 and x_2 .

3 Predecessor, Reachable, and Controllable Sets (20 points)

Consider the scalar discrete-time system

$$x_{k+1} = x_k + u_k,$$

with state constraint

$$x_k \in X = [-2, 2],$$

input constraint

$$u_k \in U = [-0.5, 0.5],$$

and target set

$$X_f = [-0.25, 0.25].$$

Please answer the following questions:

- (a) Compute the one-step predecessor set of X_f :

$$\text{Pre}(X_f) = \{x \in X : \exists u \in U \text{ such that } x + u \in X_f\}.$$

- (b) Compute the two-step controllable set to X_f :

$$K_2 = \text{Pre}(\text{Pre}(X_f)).$$

- (c) Compute K_3 . What pattern do you observe?

- (d) Starting from the initial set

$$X_0 = [-0.25, 0.25],$$

compute the one-step and two-step reachable sets under $u_k \in U$, while enforcing the state constraint X .

- (e) Compare the meaning of the controllable set in parts (a)–(c) with the reachable set in part (d). Why are they conceptually different, even though both are computed using the system dynamics?

4 Invariant Sets for an Autonomous System (20 points)

Consider the autonomous system

$$x_{k+1} = ax_k,$$

with constraint

$$x_k \in X = [-1, 1].$$

For each value of a , determine whether X is positively invariant. If it is not invariant, find the largest interval of the form $[-\alpha, \alpha] \subseteq X$ that is positively invariant.

- (a) $a = 0.5$
- (b) $a = 1$
- (c) $a = 1.2$
- (d) Based on your answers, explain the relationship between stability of the autonomous system and invariance of the constraint set.

5 Control Invariant Sets (20 points)

Consider the scalar system

$$x_{k+1} = 1.2x_k + u_k,$$

with constraints

$$x_k \in X = [-2, 2], \quad u_k \in U = [-0.3, 0.3].$$

A set $S \subseteq X$ is control invariant if for every $x \in S$, there exists $u \in U$ such that

$$1.2x + u \in S.$$

Please answer the following questions:

- (a) Show that $X = [-2, 2]$ is not control invariant.
- (b) Suppose $S = [-\alpha, \alpha]$. Derive the condition on α such that S is control invariant.
- (c) Find the largest α such that $S = [-\alpha, \alpha]$ is control invariant.
- (d) For $x = 1.5$, determine whether there exists an admissible input $u \in [-0.3, 0.3]$ that keeps the next state inside the maximal interval found in part (c).

6 Computing Controllable Sets Using MPT3 (Bonus Question: 20 points)

Consider the system

$$x_{k+1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} u_k,$$

with state constraints

$$X = \{x : -5 \leq x_1 \leq 5, -2 \leq x_2 \leq 2\},$$

input constraint

$$U = \{u : -1 \leq u \leq 1\},$$

and target set

$$X_f = \{x : -0.5 \leq x_1 \leq 0.5, -0.5 \leq x_2 \leq 0.5\}.$$

Please answer the following questions:

- (a) Use MPT3 to compute and plot the one-step, two-step, and three-step controllable sets to X_f .
- (b) Plot the sets on the same figure. Use different colors or transparencies.
- (c) Explain why the N -step controllable set generally grows as N increases.
- (d) Does the controllable set eventually stop growing? Explain what it means if

$$K_{N+1} = K_N.$$